

1. Background of the Organization

National Election Board of Ethiopia is the governmental organization responsible for organizing, managing, and supervising elections in Ethiopia. The organization ensures that elections are conducted fairly, transparently, and according to the constitution and electoral laws of the country. Its responsibilities include voter registration, candidate registration, election scheduling, vote collection, result management, and announcement of election outcomes.

Currently, many election activities rely heavily on manual or semi-manual processes. These traditional methods can consume a large amount of time, require extensive human resources, and may increase the possibility of errors, delays, and security problems. As the number of voters continues to grow, there is a need for a modern computerized system that can improve efficiency, accuracy, transparency, and security in the election process.

To address these challenges, the proposed Electronic Voting System is designed to help the National Election Board of Ethiopia manage election activities digitally. The system will provide secure voter management, electronic vote casting, automatic vote counting, and fast result generation while reducing paperwork and operational complexity.

2. Problem Statement

The current election management process used by the National Election Board of Ethiopia faces several challenges due to its dependence on manual procedures and paper-based operations. These challenges negatively affect the efficiency, transparency, and reliability of the election process.

Some of the major problems include:

- Long waiting times during voter registration and voting processes.
- Human errors during vote counting and result compilation.
- Difficulty in managing large amounts of voter data manually.
- Risk of duplicate voting and voter fraud.
- Delays in announcing election results.
- High operational costs due to paper usage and manual labor.
- Limited data security and difficulty in maintaining accurate records.
- Difficulty in generating fast and reliable election reports.

Because of these problems, there is a need to design a secure and efficient Electronic Voting System that can automate election processes, improve data management, increase transparency, and provide accurate election results in a short period of time.

3. Objectives of the Project

3.1 General Objective

To design and develop a secure, reliable, and efficient Electronic Voting System for the National Election Board of Ethiopia that improves election management and ensures accurate and transparent voting processes.

3.2 Specific Objectives

The specific objectives of the system are:

- ✓ To create a database system for storing and managing voter information securely.
- ✓ To automate the voter registration process.
- ✓ To enable secure electronic vote casting for registered voters.
- ✓ To prevent duplicate voting and unauthorized access.
- ✓ To automate vote counting and result calculation.
- ✓ To generate accurate election reports quickly.
- ✓ To improve transparency and efficiency in the election process.
- ✓ To reduce paperwork and manual operational costs.
- ✓ To maintain the confidentiality and integrity of election data.
- ✓ To provide backup and recovery mechanisms for election records.